

Kubernetes Task – Sudir Pandian

Task Description:

Setup minikube at your local and explore creating namespaces (Go through official documentation).

Techstacks needs to be used :

- Vbox, WSL
- Docker
- Minikube
- Kubectl

If the Local system has any issues you can use AWS.

- AWS EC2 (t2.medium)

How do I submit my work?

- Push all your work files to GitHub (O/P screenshot images must).
- Submit your URLs in the portal.

Terms and Conditions?

- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

Step 1: Install kubectl

```
curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl -s https://storage.googleapis.com/kubernetes-release/release/stable.txt)/bin/linux/amd64/kubectl
```

```
chmod +x kubectl
```

```
sudo mv kubectl /usr/local/bin/
```

```
root@learninglinux:~# curl -LO https://storage.googleapis.com/kubernetes-release
/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 53.7M  100 53.7M    0     0  5661k      0  0:00:09  0:00:09 --:--:-- 6607k
root@learninglinux:~# chmod +x kubectl
root@learninglinux:~# ll
total 55120
drwx-----  4 root root    4096 Mar 17 10:34 ./
drwxr-xr-x 20 root root    4096 Feb  9 10:05 ../
drwxr-xr-x  2 root root    4096 Feb 17 06:40 .aws/
-rw-----  1 root root    4399 Mar  6 11:08 .bash_history
-rw-r--r--  1 root root    3106 Apr  9 2025 .bashrc
-rwxr-xr-x  1 root root 56381592 Mar 17 10:34 kubectl*
-rw-----  1 root root      20 Feb 10 10:55 .lessht
-rw-r--r--  1 root root    132 Apr  9 2025 .profile
-rw-r--r--  1 root root     75 Feb  9 10:29 .selected_editor
drwx-----  2 root root    4096 Feb  9 10:11 .ssh/
-rw-----  1 root root   10720 Mar  6 10:59 .viminfo
root@learninglinux:~# sudo mv kubectl /usr/local/bin/
```

Step 2: Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
```

```
chmod +x minikube-linux-amd64
```

```
sudo mv minikube-linux-amd64 /usr/local/bin/minikube
```

```

root@learninglinux:~# curl -LO https://storage.googleapis.com/kubernetes-release/release/$(curl
/bin/linux/amd64/kubectl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 53.7M  100 53.7M    0     0  5661k      0  0:00:09  0:00:09 --:--:-- 6607k
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-rw-----  1 root root     10720 Mar  6 10:59 .viminfo
root@learninglinux:~# sudo mv kubectl /usr/local/bin/

```

Step 3: Start Minikube Cluster

minikube start --driver=docker --force

```

root@learninglinux:~# minikube start --driver=docker --force
* minikube v1.38.1 on Ubuntu 25.04 (vbox/amd64)
! minikube skips various validations when --force is supplied; this may lead to unexpected behavior
* Using the docker driver based on user configuration
* The "docker" driver should not be used with root privileges. If you wish to continue as root, use --force.
* If you are running minikube within a VM, consider using --driver=none:
*   https://minikube.sigs.k8s.io/docs/reference/drivers/none/
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.

X The requested memory allocation of 3072MiB does not leave room for system overhead (total system memory: 3403M)
* Suggestion: Start minikube with less memory allocated: 'minikube start --memory=3072mb'

* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
  > preloaded-images-k8s-v18-v1....: 272.45 MiB / 272.45 MiB 100.00% 4.99 Mi
  > gcr.io/k8s-minikube/kicbase....: 519.58 MiB / 519.58 MiB 100.00% 6.44 Mi

* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.35.1.
  - Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

```

```

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```

Cluster Verification

kubectl get nodes

```
root@learninglinux:~# kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
minikube      Ready     control-plane  67s   v1.35.1
```

View Default Namespaces

kubectl get ns

```
root@learninglinux:~# kubectl get ns
NAME                STATUS    AGE
default             Active    77s
kube-node-lease     Active    77s
kube-public         Active    77s
kube-system         Active    77s
```

Create Namespaces

kubectl create ns dev

kubectl create ns test

```
root@learninglinux:~# kubectl create ns dev
namespace/dev created
root@learninglinux:~# kubectl create ns test
namespace/test created
```

Verify Namespaces

kubectl get ns

```
root@learninglinux:~# kubectl get ns
NAME                STATUS    AGE
default             Active    11m
dev                 Active    10m
kube-node-lease     Active    11m
kube-public         Active    11m
kube-system         Active    11m
test                Active    4s
```

Deployment in Namespace

Create Deployment in dev Namespace

kubectl create deployment nginx --image=nginx -n dev

```
root@learninglinux:~# kubectl create deployment nginx --image=nginx -n dev
deployment.apps/nginx created
```

Check Pods in Namespace

kubectl get pods -n dev

```
root@learninglinux:~# kubectl get pods -n dev
NAME                                READY   STATUS    RESTARTS   AGE
nginx-56c45fd5ff-qmthx             1/1     Running   0           13m
```

Switch Default Namespace

kubectl config set-context --current --namespace=dev

kubectl get pods

```
root@learninglinux:~# kubectl config set-context --current --namespace=dev
Context "minikube" modified.
```

```
root@learninglinux:~# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-56c45fd5ff-qmthx             1/1     Running   0           13m
```

Delete Namespace

kubectl delete ns test

```
root@learninglinux:~# kubectl delete ns test
namespace "test" deleted
root@learninglinux:~# kubectl get ns
NAME                STATUS    AGE
default             Active   16m
dev                 Active   15m
kube-node-lease     Active   16m
kube-public         Active   16m
kube-system         Active   16m
root@learninglinux:~#
```